UNIT 1

BASIC OF PROGRAMMING

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TOPICS

1) Overview of C programming
2) Control Structures
3) Array, structures & functions
4) Basic of object oriented programming
5) Introduction to data structures.

C is a programming language developed at AT & T's Bel laboratory of USA in 1972. It was designed and written by a man named Denni's Ritchie

Steps in learning English language

Alphabets — Words — Sentence — Paragraph

Steps in learning C

Alphabets, Digits, Special Symbol — Constant, Variables, Keywords — Instructions — Programmin

C Constant
A constant is an entity that doesn't change.

C Variable
A variable is an entity that may change.

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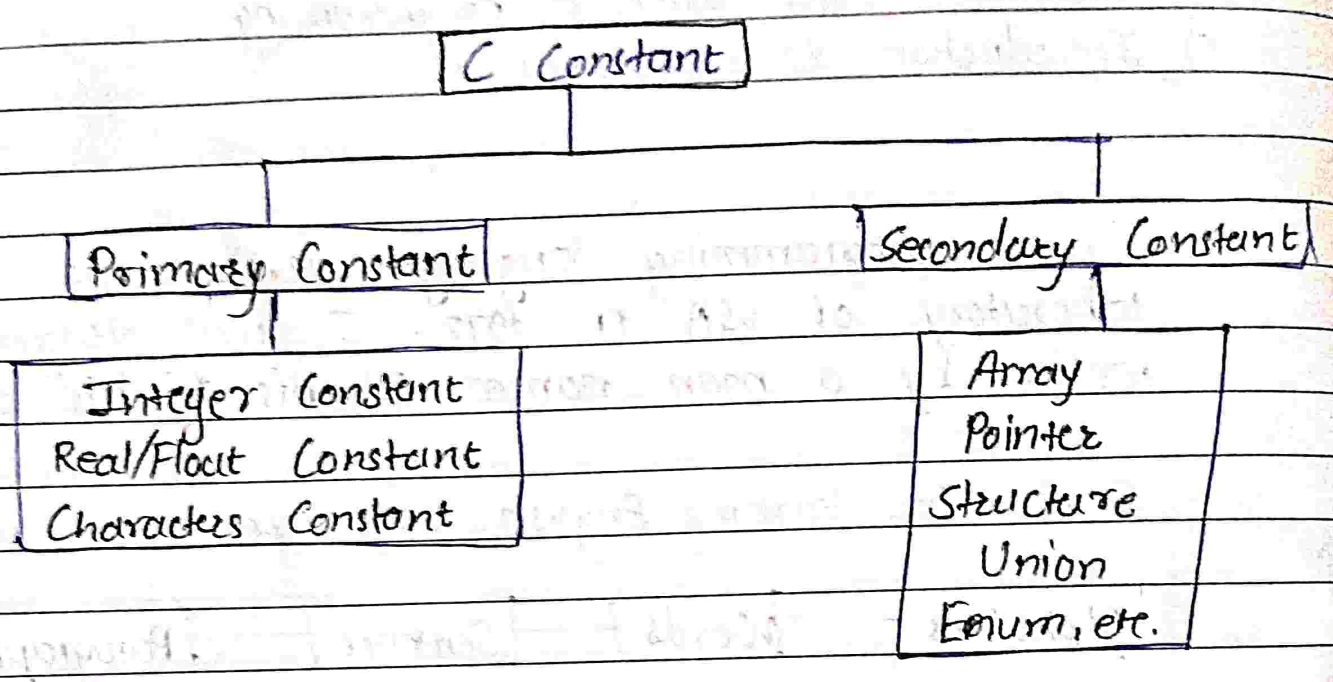
Types of C constants

The C constant are divided into two types:

(i) Primary

(ii) Secondary



Rules for constructing integer Constant

i) An integer must have atleast one digit.

ii) It must not have decimal point

iii) It can be either positive or negative

iv) If no sign preceeds an integer constant, it is

v) assumed to be positive

vi) No commas or blanks are allowed within an integer constant

vii) The allowable range for integer constant is - 32768 to 32768

The range of an integer constant depends upon the compiler.

For 16-bit compiler like turbo C & turbo C++ the range is -32768 to 32768

Ex: For intg. constant

426, +786, -8000, -760

Rules for constructing Real / Float Constant

i) A real constant must have atleast 1 digit

ii) It must have decimal point

iii) It could be either positive or negative

iv) If no sign precedes a real constant, it is assumed to be +ve

v) No commas or blanks are allowed within real constant

Ex: +325.94, 406.0, -32.76, -48.5792

Rules for constructing character constant.

i) A character constant is a single alphabate, a single digit or single special symbol enclosed within single inverted comma. Both the inverted comma should point to the left.

Ex: 'A' → Valid

'A' → Invalid

ii) The maximum length of a character constant can be one character

Ex: 'S' 'K'

C Variable

While C programming we are using an entity, the entity is which will change during the programming is called as variable.

Variables names are given to the location in memory. This location can contain integer, real or character constant. The variable and constants can used in programming. Type of variable can support type of constant.

Rules of constructing variable

i) A variable name is any combination of 1 to 31 alphabet, digits, or underscore some compiler allows name whose length could se upto 247 characters. Do not create a long variable names as it adds your typing efforts

ii) The first character in the variable name must be alphabet or underscore

iii) No commas or blanks are allowed.

iv) No special symbols other than underscore can be used in a variable name

eg: si_nt, pop_e_89, m_hra

Keywords

Keywords are the words meaning has already explained to the C computer.

The keyword cannot be used as a variable name. Because they keywords and variables getting mixed together.

There are only 32 keywords available in C

auto	double	int	struct
break	else	iong	switch
case	enum	register	type of
char	exterm	resturn	union
const	float	short	un-signed
continue	for	signed	void
default	go to	size of	volatile
do	if	static	while

main ( )

Main ( ) is a function. A function is nothing but a set of statement in C program. There can be a multiple function. The name of this function has to be main( ). All the statement than belong to main( ) are enclosed with a pair of brackets { } as shown

eg:-

```
#Include <stdio.h>
main( )
{
    Statement 1;
    statement 2;
    statement 3;
}
```

C - Instructions

There are basically 3 types of C instructions:

(i) Type Declaration Instruction

(ii) Arithematic Instruction

(iii) Control Instruction.

(i) Type Declaration Instruction

This instruction is used to decide declare the type of variable used in a C program.

Any variable used in a program must be declared before using it in any statement

The type of declaration statement is written at the beginning of main( ) function.

Ex: int. bas ;

Float rs. ; gross sal ;

char name , cod ;

(ii) Arithematic Instruction

A C arithematic Instruction consist of variable name on left hand side of = variable name & constant at R.H.

Ex: int ad ;

Float Kot , deta , alpha , beta , gamma

ad = 3200

Kot = 0.0055

deta = alpha * beta / gamma + 32 * 2/5 ;

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(iii) Control Instructions:
Control Instruction enable to specify the order in which the various instruction in a program are to be executed by the computers.

OR

The control instruction determine the flow of control in a program.

There are 4 types of control instructions used in C
(a) Sequence Control Instruction
(b) Selection or Decision Control Instruction
(c) Repitition or loop Control Instruction
(d) Case - Control Instruction.

OPERATORS / OPRAND'S
The variables and constant together are called oprand that are operated upon by the arithmetic operator and the result is assigned, using assignment operators to the variables on LHS.

Arithmetic Operators
The operators or symbol which are use to perform arithemetic / mathematical operation are called arithmetic operators. Ex: +, -, *, /, %

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Relational Operators

Relational operator are popularly used for comparision purpose. These operators check the relation between the values of variable

Sr. No	Operator	Meaning
1	<	less than
2	>	greater than
3	<=	less than equal to
4	>=	greater than equal to
5	=	equal to
6		Not equal to

Logical Operator

Logical Operators are used to evaluate the conditions or expressions. The logical operators are AND, OR & NOT.

The AND operator checks for all the condition to be true then only it returns true else false.

Sr. No.	Operator	Meaning
1	AND &&	AND-true if all condit^n are true
2	OR ||	OR-true if any 1 all cond^n are true
3	NOT !	NOT-negation.

Ex: C = a+b

Expression is a set of operator & opprands. Consider the above expression, here C will store add^n of A&B

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Assignment Operators

The assignment operators are used regularly is "=".
This is used to assign some value to the variables.
Ex: b=10, Here we are assigning the value 10 to the variable b.

C language use some short hand operators to assign the values.
Ex: a* = 7 ; b+ = 5

Increment or Decrement OPERATORS

Increment & Decrement operators are the shortcuts to increment the value.
"++" operator is used to increment/increase the value by 1.
"--" operator is used to decrement/decrease the value by 1.

Ex: ++ x is similar to x = x+1 - If x=5 then ++x=6
-- x is similar to x = x-1 If x=5 then --x=4

Conditional Operators

Conditional operator is combination of "?". This is called as ternary operation.
The general for expression 1? expression 2? expression
It checks for the condition & evaluates the results depending on the status of condition i.e. true / False

Ex: Max = a>b ? a:b

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Bitwise Operator

Bitwise operator are used to manipulate the data at bit level. Basically computer does not understand any language it understands machine codes which is in the form 0's & 1's

Operator	Meaning
&	Bitwise AND
|	Bitwise OR
^	Bitwise Exclusive OR
<<	Shift left
>>	Shift right
~	Bitwise NOT

Formatted Input / Output

Input & Output are the processes of accessing data & displaying data or results from & to the users.

- These operation are done by an operating system.

C has readymade standard library function to accept and display the data. The simplest input mechanism is to read 1 character at a time from standard input "getchar ( )".

Everytime the "getchar( )" function is called, it waits till key is pressed & return its value.

As get char ( ) is used of input, putchar ( ) is used for output.

Formatted Input

Input is the process of accepting value from the user that is from the Keyboard.
Formatted input data is an input data which has been arranged in a particular format.

Ex: Techmax C programming 1 165.50

In the above line 4 pieces of data is arranged in a specific format.
The data used by C language consists of character, number or digit or symbol. To accept all these value using a single statement "scanf( )" function is available. scanf( ) stands for scan formatted
Syntax:

scanf("control string", arg1, arg2, arg3, --- argn)

Formatted Output

The print f( ) function is used to print messages and numerical value of variable.
Using print f( ) function the printing becomes highly effective. The general form of print f( ) function or statement is as follows:

print f("control string", arg1, arg2, arg3, --- argn)

The control string in the formatted output is consist of 3 main parts:

1. Character or string or message that would be most likely to print on the screen.

(ii) The format specifiers or data specifiers to print the value of variable like %, %s

(iii) Escape sequence or back slash character such as \n, \b, \t to display the value of the output in desired format.

Identifiers

Identifiers in a C language represents the names of various entities such as array, function, variables, user defined data types labels etc.

An identifier is a type of string of alpha numeric character that always begins with either an alphabetic or underscore character.

Data Types

The data types is a collection of data which values having fixed values meaning as well as its characteristics:

The data type in C ~~are~~ can be as follows:

(i) Character

(ii) Integer

(iii) Floating point arithematic

(iv) Double precision floating point format.

* Character : Character data type is used to store single character or no. ~~of~~ at a time. The memory space required for character type of data is 8-byte for 16-bit machine

Range of values is -128 to 127.

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* Integer : Integer allows to store integer values only. No decimal points are allowed. Integer takes 16 bytes of memory to store the values on 16- bit machines.
The range of values is -32768 to +32767

* Floating Point : Floating type store only floating point value The values should be with the decimal places. Float types takes 32 bytes of memory to store floating values on 16 - bit machines. Range of values is $3.4e^{-38}$ to $3.4+e^{38}$

* Double Precision: Double types stores the double value than float. Double type takes 64 bytes of memory to get stored on 16 - bit machine.
Range is $1.7\ e^{-308}$ to $1.7e^{+308}$

Types	Size (bits)	Range
Signed char	8	-128 to 127
Unsigned char	8	0 to 255
signed int	16	-32.768 to 32767
unsigned int	16	0 to 65535
short int or unsigned short int	8	-128 to 127
long int or signed long int	8	0 to 255
signed long int	32	-2,147,483,648 to 2147,483,647
unsigned long int	32	0 to 4,294,967,295
float	32	$3.4\ E^{-38}$ to $3.4\ E^{+38}$
Double	64	$1.7\ E^{-308}$ to $1.7E^{+308}$
long Double	80	$3.4\ e^{-4932}$ to $1.1\ e^{+4932}$

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COMPILER

A compiler is a special program that translates a programming language source code into machine code or another programming language.

The source code is typically written in a high level language, human readable language such as Java or C++.

OR

Compiler is use to allow you to write a C code and run it online.

Decision making is used to make a particular decision regarding slecting a particular language.

In a C language the control statements are used to control the program. Control statements are designed / for by following 3 parameters:

① Selection ② Repitation ③ Sequence

The control statements which are used in C are as follows:

① If - else
② Switch - case
③ For loop
④ While loop
⑤ do while loop

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Decision making using If

Simple If statement can be used to check wether condition is true or not. In if statement only true part is included. The first loop for taking decision is If.

The syntax ~~ye~~ for representing simple If is

```
If [condition]
{ - - - - - - -
  - - statement
  - - - - - - -
}
```

eg:-

```
if (x=10)
{ ~~y=x+5~~
  y= x+5
}
```

In above example if (x=10) then y will be come x+5 This is called simple 'if' statement.

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